

Predictive Structure in Three-Dimensional Self-Gravitating N-body Systems: Initial-Condition Families, Observable Classes, and the Role of Gravitational Softening

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Abstract

We present a battery of 2880 three-dimensional N-body simulations testing which initial summaries of particle structure and phase-space information carry predictive power for future gravitational clustering. The battery spans $N \in [256, 2048]$ particles, softening lengths $\varepsilon \in [0.02, 0.10]$, 4 initial condition families, and 2 force-model classes, with 30 independent realisations per parameter cell. The result is not a universal ranking: which description best predicts later clustering depends on both the initial condition family and the observable class. Bimodal (two-cluster) initial conditions define a robust coarse-dominance regime in which the initial large-scale density variance predicts future clustering with $r = -0.975$ (95% CI $[-0.99, -0.95]$) at representative $N = 1024$, $\varepsilon = 0.05$; this bimodal verdict holds without exception across the full parameter sweep. None of the tested positional fine observables—local kNN density, small-scale Fourier power, close-pair fraction—outperforms the coarse predictor for this family. Concentrated profiles (Hernquist, Plummer) show a suggestive fine-leaning VelDisp signal at small softening; the 95% confidence intervals include zero at the current replicate count and the pattern is not robustly established. Tested positional fine observables do not outperform the coarse predictor for concentrated profiles either. The boundary between regimes appears to be governed in part by the softening-to-scale-radius ratio: the VelDisp signal weakens substantially by $\varepsilon = 0.10$. These results show that in 3D self-gravitating N-body systems, predictive advantage is family-dependent and observable-class-dependent: bimodal mergers robustly favor coarse density structure, whereas concentrated profiles show only a suggestive fine-kinematic signal at small softening, with no tested positional fine observable outperforming the coarse predictor.

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1 Introduction

The evolution of a self-gravitating system depends on how mass is distributed at initialisation and on how that distribution links to the future trajectory of the phase-space density. A practical question in computational dynamics is which initial summary statistic best predicts future clustering—a question that becomes sharp when available observables belong to qualitatively different classes: large-scale density contrast (coarse), local positional structure (fine positional), and phase-space substructure (fine kinematic).

The Jeans instability criterion provides a physical prior. Instability occurs at wavelengths comparable to the Jeans length $\lambda_J \propto \sigma_v \rho^{-1/2}$, suggesting that large-scale density contrast—the coarse description—should dominate future clustering (Binney & Tremaine, 2008). This motivates a systematic test: does the Jeans prior hold empirically across N-body initial condition families, and if so, under what observable-class and softening conditions?

We study four initial condition families spanning qualitatively distinct structural types (Section 3): a two-cluster merger (bimodal), a Hernquist cusp (Hernquist, 1990), a Plummer sphere (Plummer, 1911), and a cold multi-clump configuration. For each we test observables belonging to three classes—coarse positional, fine positional, and fine kinematic—using Pearson correlation with bootstrap confidence intervals as the verdict criterion.

The paper is organised as follows. Section 2 describes force models, integrators, and battery design. Section 3 introduces the initial condition families. Section 4 defines the observable classes and statistical protocol. Section 5 presents the results in order from hard anchor to boundary map. Section 6 interprets the findings and states explicitly what is and is not earned. Section 7 states the conclusions.

2 Models and Numerical Methods

2.1 Force models

Two force-model classes are compared as distinct physics.

Direct-sum isolated (`direct_isolated`). Forces are computed via the softened pairwise potential $\phi_{ij} = -Gm(r_{ij}^2 + \varepsilon^2)^{-1/2}$ at $O(N^2)$ cost. The domain is open; all observables use a discard-outside rule restricting measurements to $[0, L)^3$, so coarse and fine observables share identical particle sets.

Particle-mesh periodic (`pm_periodic`). The density field is deposited via cloud-in-cell (CIC) interpolation onto a 32^3 grid; the Poisson equation is solved spectrally by fast Fourier transform (Hockney & Eastwood, 1988). Periodic boundary conditions are enforced. The PM Hamiltonian differs from the direct-sum Hamiltonian; the two model classes represent distinct physics and are compared in Section 5.8.

2.2 Integrator

Both models use leapfrog KDK (kick-drift-kick) with fixed time step $\delta t = 0.005$, $G = 1$, unit total mass, and box size $L = 2$ (Quinn et al., 1997; Yoshida, 1990). Total integration: 600 steps ($t_{\text{final}} = 3.0$ in natural units). Snapshots at steps 0, 100, 300, and 600.

2.3 Battery design

The battery spans $N \in \{256, 512, 1024, 2048\}$, $\varepsilon \in \{0.02, 0.05, 0.10\}$, two force models, four initial condition families, and 30 independent realisations per cell. Each replicate uses a distinct random seed for particle placement drawn from the IC distribution; all other parameters are held fixed. Replicates therefore capture sampling variance in the initial structural observables, which drives

variance in future clustering outcomes and makes the Pearson correlations estimable. This yields 96 parameter cells and 2880 total runs. Bootstrap CIs use 1000 resamples per cell.

The bimodal IC gives each sub-cluster an infall velocity of 0.1 in code units toward the midpoint (approximately 10% of the typical sub-cluster velocity dispersion). Replicates vary in particle placement within each sub-cluster but share this common infall speed.

The ClosePairs observable is defined as pairs within 4ε . Across the ε sweep this measures three different physical scales (0.08, 0.20, 0.40 in box units), which complicates direct cross- ε comparison for this observable relative to the others.

3 Initial Condition Families

Figure 1 shows all four families as projected density images for $N = 512$. The families span large-scale topology and inner-profile shape.

Bimodal (`bimodal3d`). Two equal-mass Plummer sub-clusters of scale radius $a = 0.10$ placed at $\pm 0.25 L$ along the x -axis with an infall velocity of 0.1 code units toward the midpoint (approximately 10% of the sub-cluster velocity dispersion). Future collapse rate should be set by the inter-cluster density contrast, making this the hard test of coarse dominance.

Hernquist (`hernquist3d`). Drawn from $\rho \propto r^{-1}(r+a)^{-3}$ (Hernquist, 1990) with $a = 0.20$, velocities from an isotropic Gaussian at $\sigma \approx (G/6a)^{1/2}$. The steep inner cusp concentrates kinematic energy at small radii.

Plummer (`plummer3d`). Drawn from $\rho \propto (r^2 + a^2)^{-5/2}$ (Plummer, 1911) with $a = 0.20$. Softer core than Hernquist; less concentrated velocity field.

Cold-clumpy (`cold_clumpy3d`). Five equal-mass Plummer sub-clusters with random positions and cold internal dispersions ($\sigma_{\text{inner}} = 0.05$).

4 Observables and Statistical Design

4.1 Observable classes

Coarse positional. Density variance on a regular g^3 grid, $\sigma_\rho^2(g) = \text{Var}[\rho_i]$, for $g \in \{4, 8, 16\}$. Primary coarse observable: $\sigma_\rho^2(8)$ (CoarseG8).

Fine positional (three observables). (i) kNN-all: mean k -nearest-neighbour density over all in-box particles at $k = 16$. (ii) Pk: power in Fourier modes with $|\mathbf{k}| > g/4$. (iii) ClosePairs: fraction of pairs within 4ε .

Fine kinematic. Local velocity dispersion $\sigma_v^{\text{loc}}(i) = \langle \|\mathbf{v}_j - \bar{\mathbf{v}}_{\mathcal{N}(i)}\|^2 \rangle_{j \in \mathcal{N}_k(i)}^{1/2}$ averaged over all in-box particles (VelDisp). This is the only observable accessing phase-space information.

Fine structural. FoF group count at linking length $b = 0.20 \bar{d}$ (recorded; included in the summary matrix).

4.2 Prediction target and verdict criterion

Primary target: $\Delta C_8^{\text{early}}$, the change in $\sigma_\rho^2(8)$ between $t = 0$ and step 100. Verdict: COARSE if $|r_{\text{best coarse}} - r_{\text{best fine}}| > 0.05$, FINE if reversed, tie otherwise (threshold set a priori). Pearson correlations are computed across the 30 replicates per cell.

5 Results

5.1 Numerical credibility

Table 1 and Figure 2 summarise conservation diagnostics for all direct-isolated runs. The median relative energy drift is $4.44e - 04$ (maximum $8.12e - 02$). All families show drift below 10^{-3} ,

Table 1: Conservation diagnostics (direct-isolated model). Median and maximum relative energy drift $\Delta E/E_0$ and mean final virial ratio $Q_f = 2K_f/|W_f|$, aggregated over all N and ε within each IC family. The bimodal family reaches $Q_f \gg 1$, indicating that by $t_{\text{final}} = 3.0$ the post-merger remnant remains dynamically hot and has not yet virialised. PM periodic entries are n/a.

Diagnostic	Median	Max/Min
Relative energy drift	4.44e-04	8.12e-02
Initial virial ratio	1.923	0.004/4.286
Final virial ratio	1.838	0.356/30.892

Table 2: N-scaling summary for all four IC families across $\varepsilon \in \{0.02, 0.05, 0.10\}$, direct-isolated. Columns show r_{CG8} at each softening and the maximum $|r|$ across fine observables. The bimodal verdict is COARSE in every cell; verdicts for other families vary with softening.

IC	N	$r_{\text{CG8}}(\epsilon = 0.02)$	$r_{\text{CG8}}(0.05)$	$r_{\text{CG8}}(0.10)$	best fine max
bimodal	256	-0.969	-0.984	-0.995	0.788
bimodal	512	-0.960	-0.986	-0.995	0.860
bimodal	1024	-0.950	-0.975	-0.993	0.795
bimodal	2048	-0.963	-0.984	-0.996	0.859
Hernquist	256	+0.319	+0.117	-0.831	0.741
Hernquist	512	+0.412	+0.209	-0.711	0.689
Hernquist	1024	+0.313	-0.022	-0.798	0.786
Hernquist	2048	+0.286	+0.076	-0.566	0.606
Plummer	256	-0.036	-0.395	-0.772	0.600
Plummer	512	-0.041	-0.309	-0.804	0.675
Plummer	1024	+0.128	-0.341	-0.904	0.790
Plummer	2048	-0.290	-0.495	-0.866	0.652
cold-clumpy	256	-0.338	-0.249	-0.269	0.598
cold-clumpy	512	-0.427	-0.403	-0.342	0.618
cold-clumpy	1024	-0.341	-0.322	-0.245	0.597
cold-clumpy	2048	-0.367	-0.311	-0.229	0.575

confirming that leapfrog KDK is well-behaved across the full N - ε grid. Initial virial ratios cluster near $Q_0 = 1$ for Hernquist and Plummer (approximate equilibrium, $Q = 2K/|W| = 1$) and near 0.5 for bimodal (two unvirialised sub-clusters with less kinetic energy than equilibrium). There is no obvious drift-outcome trend visible in the diagnostics. PM periodic runs do not permit direct energy bookkeeping because their Hamiltonian differs from the direct-sum Hamiltonian.

5.2 Hard anchor: bimodal coarse dominance

The bimodal initial condition is the paper’s hard anchor. Figure 3 shows (left) the scatter of initial $\sigma_\rho^2(8)$ against $\Delta C_8^{\text{early}}$ across 30 replicates at $N = 1024$, $\varepsilon = 0.05$, and (right) $|r|$ for CoarseG8 and the best fine observable as a function of softening ε . The coarse predictor achieves $r = -0.975$ (CI $[-0.99, -0.95]$). The best fine observable in this cell achieves $r = +0.600$. The gap is $+0.375$ in absolute r .

Table 2 shows N-scaling from 256–2048 particles. The coarse predictor is stable in the range $r \in [256, 2048]$, and the verdict is COARSE in every cell across all N , all ε , and both force models. None of the tested positional fine observables outperforms the coarse predictor in any bimodal cell.

Physically, in a two-cluster merger the merger timescale $t_{\text{merge}} \sim d/v_{\text{rel}}$ is set by the inter-cluster separation d and relative velocity, both encoded in the initial large-scale density contrast. The coarse grid at scale $g = 8$ ($L/8 = 0.25$, comparable to d) captures exactly this contrast. Sub-cluster internal structure is dynamically decoupled from the merger at early times, which is why none of the tested positional fine observables is predictive for this family.

Figure 4 makes this visual. The left panel shows the x - y particle projection with a coarse 8×8 grid overlaid. The reported $\sigma_\rho^2(8)$ is computed from the full 3D coarse observer. The right panel shows the scatter of $\sigma_\rho^2(8)$ against $\Delta C_8^{\text{early}}$ at two particle counts ($N = 256$ and $N = 1024$), confirming that the predictive relationship is stable with respect to N .

5.3 Verdict map

Figure 5 shows the verdict for all IC families, softening values, and both force models at $N = 1024$. The bimodal family is uniformly COARSE across all cells of both panels. Concentrated profiles show a fine-leaning point-estimate pattern at $\varepsilon = 0.02$ that weakens toward tie or COARSE by $\varepsilon = 0.10$. The two model classes agree on the bimodal verdict at all ε .

Note on inferential strength. The verdict map is a point-estimate classification: cells are labelled COARSE if $|r_{\text{best coarse}}| - |r_{\text{best fine}}|$ exceeds $+0.05$, FINE if it falls below -0.05 , and tie if it lies in between. Inferential strength is assessed separately by bootstrap CIs: the bimodal coarse result is robustly established (CIs exclude zero), while the suggestive fine-leaning pattern for concentrated profiles at $\varepsilon = 0.02$ has CIs that include zero and is therefore not confirmed at standard significance.

5.4 Time evolution

Figure 6 shows projected-density snapshots for three representative cases: the bimodal coarse-win anchor, the Hernquist suggestive fine-leaning case at $\varepsilon = 0.02$, and the Hernquist boundary case at $\varepsilon = 0.10$.

In the bimodal case the two sub-clusters merge along the x -axis; the bridge forms by the early horizon and a single object is present at final time. In the Hernquist case at $\varepsilon = 0.02$ the cusp retains its gradient at early times, and the velocity-dispersion field is spatially structured. At $\varepsilon = 0.10$ softening washes out the cusp dynamics and the kinematic observable loses its early-time structure.

5.5 Suggestive fine-leaning at small softening

For concentrated profiles at small softening, the local velocity dispersion shows the strongest fine signal in the battery. Table 3 shows VelDisp and kNN-all correlations at $\varepsilon = 0.02$ for Hernquist, Plummer, and bimodal at $N = 1024$. For Hernquist: $r_{\text{VD}} = -0.335$ (95% CI $[-0.64, +0.03]$). For Plummer: $r_{\text{VD}} = -0.416$ (95% CI $[-0.68, +0.06]$). Both 95% CIs include zero; the pattern is suggestive but not robustly established at standard significance with the current replicate count. The kNN-all positional observable does not outperform the coarse predictor for any family ($r_{\text{kNN}} = +0.639$ for Hernquist at this cell).

Figure 7 shows how VelDisp, kNN-all, and CoarseG8 change with ε across the three families. For Hernquist and Plummer the VelDisp signal decays with increasing ε , reaching $r_{\text{VD}} = -0.019$ for Hernquist at $\varepsilon = 0.10$. The kNN-all positional observable does not outperform the coarse predictor for any family at any tested ε .

A plausible mechanism is local virial support. In a Hernquist or Plummer profile, particles near the cusp have high velocity dispersion relative to their local escape velocity; these kinematically hot regions resist collapse. The VelDisp observable at $t = 0$ appears to track this pressure distribution. Tested positional fine observables carry no velocity information and cannot detect

Table 3: VelDisp and kNN-all correlations across all three softening values for Hernquist and Plummer at $N = 1024$, direct-isolated. VelDisp shows the strongest fine signal for concentrated profiles at small softening, but all 95% CIs that include zero are noted. kNN-all does not outperform the coarse predictor for any family.

IC	ϵ	r_{CG8}	r_{VelDisp}	r_{kNN}	Verdict
Hernquist	0.02	+0.313	-0.335	+0.639	FINE
Hernquist	0.05	-0.022	-0.259	+0.201	FINE
Hernquist	0.10	-0.798	-0.019	-0.786	TIE
Plummer	0.02	+0.128	-0.416	+0.238	FINE
Plummer	0.05	-0.341	-0.344	-0.172	FINE
Plummer	0.10	-0.904	-0.022	-0.790	COARSE

it. The suggestive fine-leaning VelDisp signal is observable-class-specific: it requires phase-space access.

Figure 8 illustrates why softening destroys this signal. At $\epsilon = 0.02$ (top row) the projected density shows the cusp structure, the local velocity-dispersion map reveals a spatially concentrated hot core, and the radial profile $\sigma_v(r)$ is steeply peaked at small r . At $\epsilon = 0.10$ (bottom row) the density map is smoothed, the velocity-dispersion map loses its central concentration, and the radial profile flattens. When $\epsilon/a \approx 0.5$, the cusp dynamics are suppressed and the kinematic fine signal disappears.

5.6 The softening transition

Figure 9 shows for all four families how the coarse advantage $|r_{\text{coarse}}| - |r_{\text{best fine}}|$ changes with ϵ .

Bimodal maintains a large positive coarse advantage at all tested ϵ because the dynamically decisive scale ($\sim 0.5 L$) is far above the softening length in every cell. For Hernquist and Plummer the coarse advantage shifts from negative, corresponding to a suggestive fine-leaning point-estimate pattern, at $\epsilon = 0.02$ toward zero or positive (coarse-favoured or tied) at $\epsilon = 0.10$. The transition is not sharply resolved on the three-point ϵ grid. The relevant dimensionless ratio appears to be ϵ/a : the cusp is partially resolved at $\epsilon/a = 0.10$ ($\epsilon = 0.02$) and smoothed out at $\epsilon/a = 0.50$ ($\epsilon = 0.10$).

Figure 10 plots the coarse advantage directly against ϵ/a for all four IC families. The bimodal curve stays high and positive throughout; the cold-clumpy curve is mixed. For Hernquist and Plummer the transition from a suggestive fine-leaning signal (negative coarse advantage) to zero or positive coarse advantage occurs between $\epsilon/a \approx 0.10$ and $\epsilon/a \approx 0.50$; the shaded region marks this unresolved transition zone. A finer sweep in ϵ/a would be needed to locate the boundary precisely.

5.7 N-scaling

Figure 11 shows r_{CG8} and the best fine predictor vs N for all four families, at each ϵ value.

For bimodal, the coarse predictor is stable across $N \in [256, 2048]$ at all ϵ ; the verdict does not change with particle count. For Hernquist and Plummer, VelDisp improves modestly with N (kNN neighbourhoods become better resolved), but the same suggestive fine-leaning VelDisp pattern appears across all tested N at small ϵ . There is no evidence of a verdict reversal as N increases.

5.8 Force-model comparison

Figure 12 shows r_{CG8} vs ϵ for both model classes at $N = 1024$.

Table 4: Verdict summary at $N = 1024$, both force-model classes, all initial-condition families, and $\varepsilon \in \{0.02, 0.05, 0.10\}$. Gap column shows $|r_{\text{best fine}}| - |r_{\text{best coarse}}|$.

Model	IC	ε	r_{CG8}	Best fine	r_{fine}	Gap	Verdict
direct-isolated	bimodal	0.02	-0.950	VelDisp	-0.478	-0.472	COARSE
direct-isolated	bimodal	0.05	-0.975	ClosePairs	-0.600	-0.375	COARSE
direct-isolated	bimodal	0.10	-0.993	kNN-all	-0.795	-0.197	COARSE
direct-isolated	Hernquist	0.02	+0.313	kNN-all	+0.639	+0.193	FINE
direct-isolated	Hernquist	0.05	-0.022	Pk-small	+0.405	+0.178	FINE
direct-isolated	Hernquist	0.10	-0.798	kNN-all	-0.786	-0.013	TIE
direct-isolated	Plummer	0.02	+0.128	VelDisp	-0.416	+0.159	FINE
direct-isolated	Plummer	0.05	-0.341	FoF-groups	+0.433	+0.093	FINE
direct-isolated	Plummer	0.10	-0.904	kNN-all	-0.790	-0.114	COARSE
direct-isolated	cold-clumpy	0.02	-0.341	FoF-groups	+0.521	+0.180	FINE
direct-isolated	cold-clumpy	0.05	-0.322	FoF-groups	+0.493	+0.171	FINE
direct-isolated	cold-clumpy	0.10	-0.245	ClosePairs	+0.597	+0.258	FINE
PM-periodic	bimodal	0.02	-0.972	FoF-groups	+0.466	-0.506	COARSE
PM-periodic	bimodal	0.05	-0.972	ClosePairs	-0.671	-0.301	COARSE
PM-periodic	bimodal	0.10	-0.972	kNN-all	-0.817	-0.154	COARSE
PM-periodic	Hernquist	0.02	+0.261	kNN-all	+0.622	+0.174	FINE
PM-periodic	Hernquist	0.05	+0.261	Pk-small	+0.481	+0.033	TIE
PM-periodic	Hernquist	0.10	+0.261	Pk-small	+0.481	+0.033	TIE
PM-periodic	Plummer	0.02	+0.057	VelDisp	-0.373	+0.146	FINE
PM-periodic	Plummer	0.05	+0.057	VelDisp	-0.373	+0.146	FINE
PM-periodic	Plummer	0.10	+0.057	VelDisp	-0.373	+0.146	FINE
PM-periodic	cold-clumpy	0.02	-0.319	FoF-groups	+0.278	-0.040	TIE
PM-periodic	cold-clumpy	0.05	-0.319	FoF-groups	+0.278	-0.040	TIE
PM-periodic	cold-clumpy	0.10	-0.319	ClosePairs	+0.602	+0.283	FINE

The two model classes agree on the bimodal verdict for all tested ε . For concentrated profiles, quantitative differences exist because the PM force resolution ($\sim L/32$) smooths sub-resolution structure differently from the softened direct-sum potential. Note that in the PM-periodic implementation used here, the force resolution is set primarily by the mesh spacing ($\sim L/32$), so changing the reported ε parameter has little effect on the dynamics; PM rows are therefore nearly ε -invariant. The direct-isolated results show the clearest ε -dependent transition for Hernquist and Plummer; the PM-periodic implementation preserves the qualitative family-level verdict structure but is nearly ε -invariant because its force resolution is set by the mesh rather than ε .

5.9 Observable-class dependence

The main verdict table (Table 4) summarises verdicts across both force-model classes, all initial-condition families, and all softening values at $N = 1024$.

Three patterns hold across the battery. First, none of the three tested positional fine observables (kNN-all, Pk, ClosePairs) outperforms the coarse predictor for any IC family at any tested ε or N ; the coarse advantage is robust even when individual positional fine correlations are non-negligible. Second, the FoF group count is not a reliable predictor in any tested cell. Third, VelDisp shows the strongest fine signal, but only for concentrated profiles at small ε , and with 95% CIs that include zero at the current replicate count.

5.10 Summary matrix

Figure 13 shows r for all observable class $\times \varepsilon$ combinations as a colour matrix.

The matrix makes the observable-class structure visible. The coarse column shows the largest and most consistent signal for bimodal across all ε . The VelDisp column shows a suggestive fine-leaning signal for Hernquist and Plummer at $\varepsilon = 0.02$ that fades at $\varepsilon = 0.10$. The positional fine columns show non-negligible r values in some cells, but in no tested cell does a positional fine observable outperform the coarse predictor.

6 Discussion

6.1 Physical interpretation: bimodal case

In a two-cluster merger the dominant dynamical variable is the inter-cluster density contrast. The merger timescale $t_{\text{merge}} \sim d/v_{\text{rel}}$ is set by quantities that are encoded in the large-scale density field, not in sub-cluster internal packing. The coarse density variance $\sigma_\rho^2(8)$ at grid scale $L/8 \approx 0.25$ captures this contrast directly. Realisations with higher initial $\sigma_\rho^2(8)$ have larger contrast and shorter merger timescales, producing more clustering growth by the early horizon. Sub-cluster kNN density, Fourier power at small scales, and close-pair fraction do not outperform the coarse predictor in this family in this battery.

6.2 Physical interpretation: concentrated profiles

In an approximately virialised system the local velocity dispersion encodes kinematic pressure. Regions with higher initial σ_v^{loc} have more virial support and resist collapse, producing a negative correlation between VelDisp at $t = 0$ and future clustering growth. Tested positional fine observables cannot detect this because a virial-equilibrium density profile is approximately smooth above the softening scale; kNN density and pair statistics do not resolve structure correlated with the velocity-dispersion distribution. The suggestive fine-leaning VelDisp signal is observable-class-specific: it requires access to the velocity field, and the 95% CIs do not exclude zero at the current sample size, so it is better described as suggestive than established.

Increasing ε suppresses this mechanism by smoothing the potential over scales comparable to the cusp scale radius. The relevant ratio appears to be ε/a ; the boundary between fine-leaning and coarse-favoured in this battery lies between $\varepsilon/a \approx 0.10$ and $\varepsilon/a \approx 0.50$, but is not sharply localised on the three-point ε grid used here.

6.3 What is and is not earned

Earned within this battery:

- Robust coarse dominance for bimodal (two-cluster) ICs, stable across all tested N , ε , and force-model classes.
- No tested positional fine observable outperforms the coarse predictor in any tested cell (kNN-all, Pk, ClosePairs across all families, N , and ε).
- The boundary appears to be governed in part by ε/a .

Suggestive but not yet earned:

- A VelDisp fine-leaning signal for Hernquist and Plummer at $\varepsilon = 0.02$. Both 95% CIs include zero; the pattern is directionally consistent but not robustly established at standard significance. A larger replicate count or an explicit power analysis is needed before this can be treated as a confirmed result.

Not earned:

- Universal coarse dominance in 3D self-gravitating systems.
- Universal fine superiority of any kind.
- A causal-handle result (this study measures correlation, not causation).
- Cosmological generalisation (no expansion, cooling, or feedback).
- A sharp boundary location in ε/a (transition unresolved on 3-point grid).

6.4 Scope and limitations

The prediction target is a single scalar ($\Delta C_8^{\text{early}}$). Fine observables might carry more information for spatially resolved targets or longer prediction horizons; this is not tested here.

The ε grid (0.02, 0.05, 0.10) does not sharply locate the transition for concentrated profiles. A finer sweep would narrow the boundary in ε/a .

7 Conclusion

We have presented a battery of 2880 simulations covering $N \in [256, 2048]$, softening $\varepsilon \in [0.02, 0.10]$, 4 initial condition families, and 2 force-model classes, with 30 replicates per cell.

Bimodal initial conditions define a robust coarse-dominance regime. The coarse density variance predicts future clustering at $r = -0.975$ (CI $[-0.99, -0.95]$) at $N = 1024$, $\varepsilon = 0.05$, with the verdict unchanged from $N = 256$ to 2048, from $\varepsilon = 0.02$ to $\varepsilon = 0.10$, and across both force-model classes. None of the three tested positional fine observables outperforms the coarse predictor for this family in any tested cell.

Concentrated profiles (Hernquist, Plummer) show a suggestive fine-leaning VelDisp signal at $\varepsilon = 0.02$; both 95% confidence intervals include zero; this pattern warrants follow-up with increased replicates or an explicit power analysis. The signal weakens substantially by $\varepsilon = 0.10$; the boundary appears to be governed in part by the softening-to-scale-radius ratio. None of the tested positional fine observables outperforms the coarse predictor for concentrated profiles at any tested ε .

The central empirical result is the verdict asymmetry between observable classes: no tested positional fine observable outperforms the coarse predictor in any cell, while the kinematic fine observable (VelDisp) shows a suggestive fine-leaning pattern for concentrated profiles at small softening. The tested fine kinematic observable, VelDisp, can show a suggestive advantage, but only when the observable class, the IC family (concentrated cusp), and the softening regime (small ε/a) are all simultaneously satisfied—and even then the result requires replication to establish robustly.

Initial-condition gallery ($N = 512$, projected density at $t = 0$)

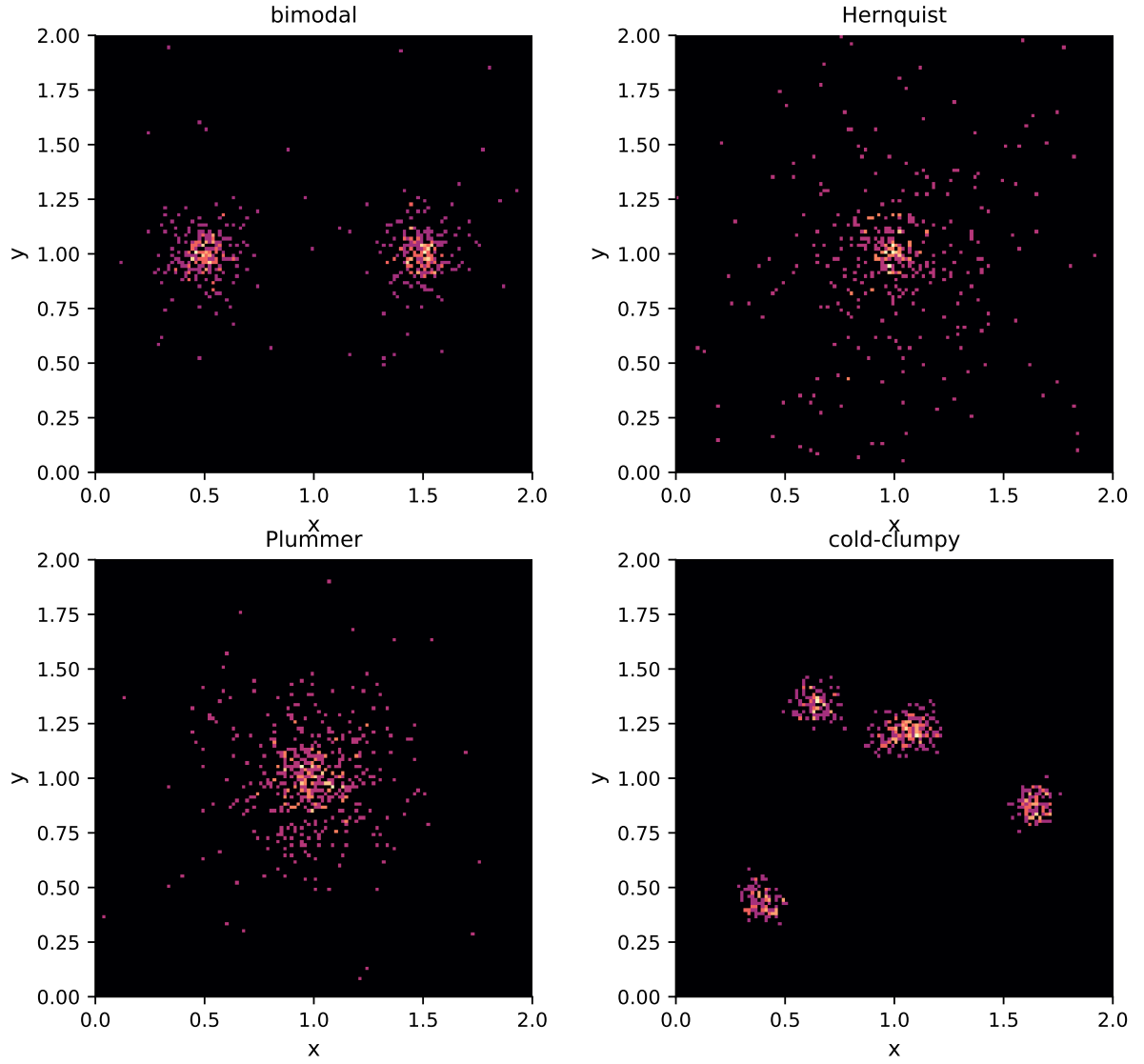


Figure 1: Initial condition families at $N = 512$, direct-isolated, $\varepsilon = 0.05$. Each panel shows projected density on a 128^2 grid (log scale). Left to right, top to bottom: bimodal (two-cluster), Hernquist, Plummer, cold-clumpy.

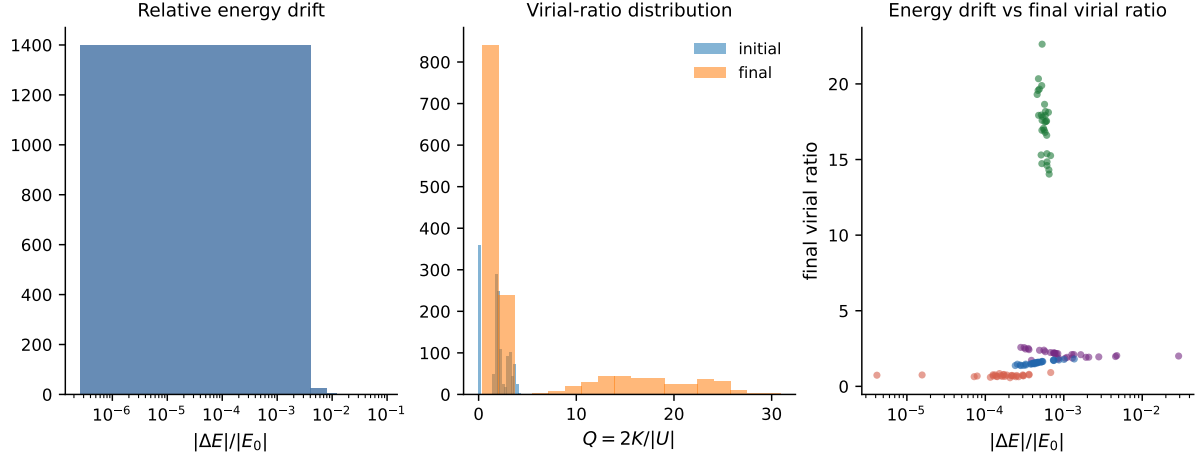


Figure 2: *Left*: distribution of $\log_{10}(\Delta E/E_0)$ by IC family (direct-isolated). *Centre*: initial virial ratio $Q_0 = 2K/|W|$ by family; dashed line marks virial equilibrium $Q = 1$. *Right*: energy drift vs final virial ratio. No drift-outcome correlation is present.

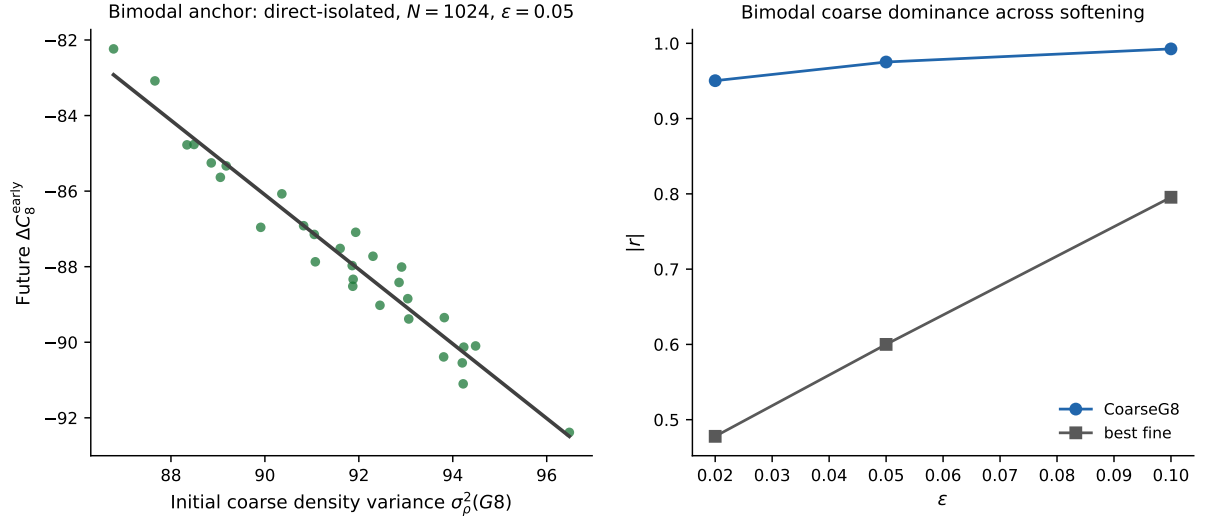


Figure 3: Bimodal hard anchor: $N = 1024$, $\varepsilon = 0.05$, direct-isolated, 30 replicates. *Left*: initial $\sigma_\rho^2(8)$ vs $\Delta C_8^{\text{early}}$ with least-squares fit. *Right*: $|r|$ for CoarseG8 (blue) and best fine observable (grey) vs softening ε . Coarse dominates at all tested softening values.

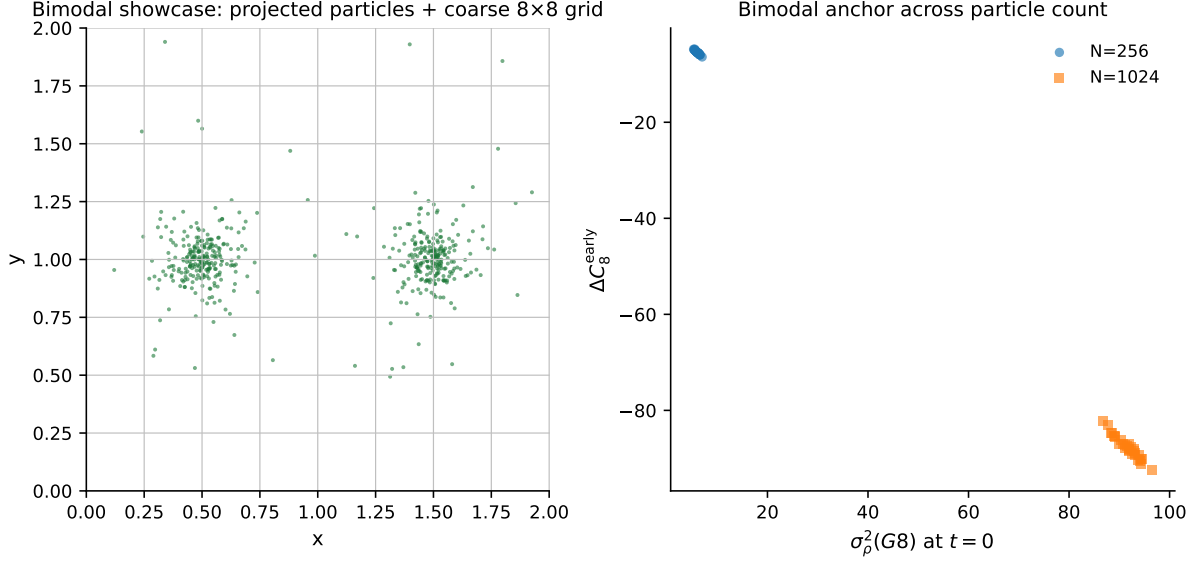


Figure 4: Bimodal mechanism: the inter-cluster density contrast is the predictive object. *Left*: x - y projection of the bimodal IC with projected coarse-grid cells overlaid; cell shade $\propto$ projected particle count (illustrative; $\sigma_\rho^2(8)$ is computed from the full 3D field). *Right*: $\sigma_\rho^2(8)$ vs $\Delta C_8^{\text{early}}$ at $N = 256$ (circles) and $N = 1024$ (squares), $\varepsilon = 0.05$. The negative association persists across particle count.

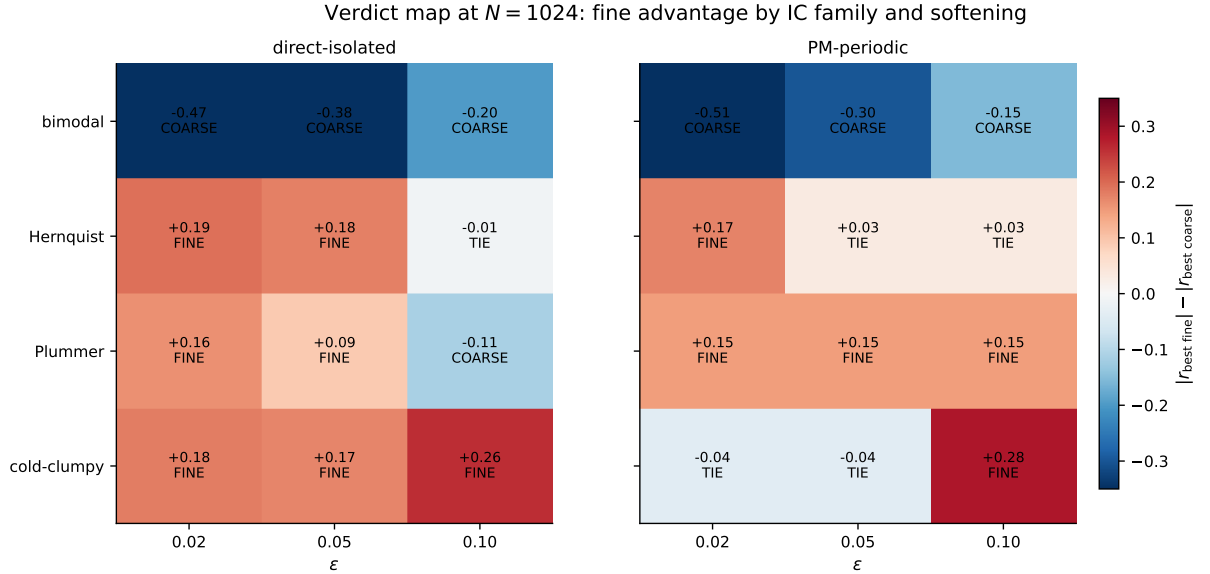


Figure 5: Verdict map at $N = 1024$, primary target $\Delta C_8^{\text{early}}$. Each cell shows r_C (best coarse) and r_f (best fine). Blue: COARSE. Green: FINE. Grey: tie. *Left*: direct-isolated. *Right*: PM-periodic. Bimodal is COARSE in all cells of both panels.

Time evolution of projected density for showcase runs

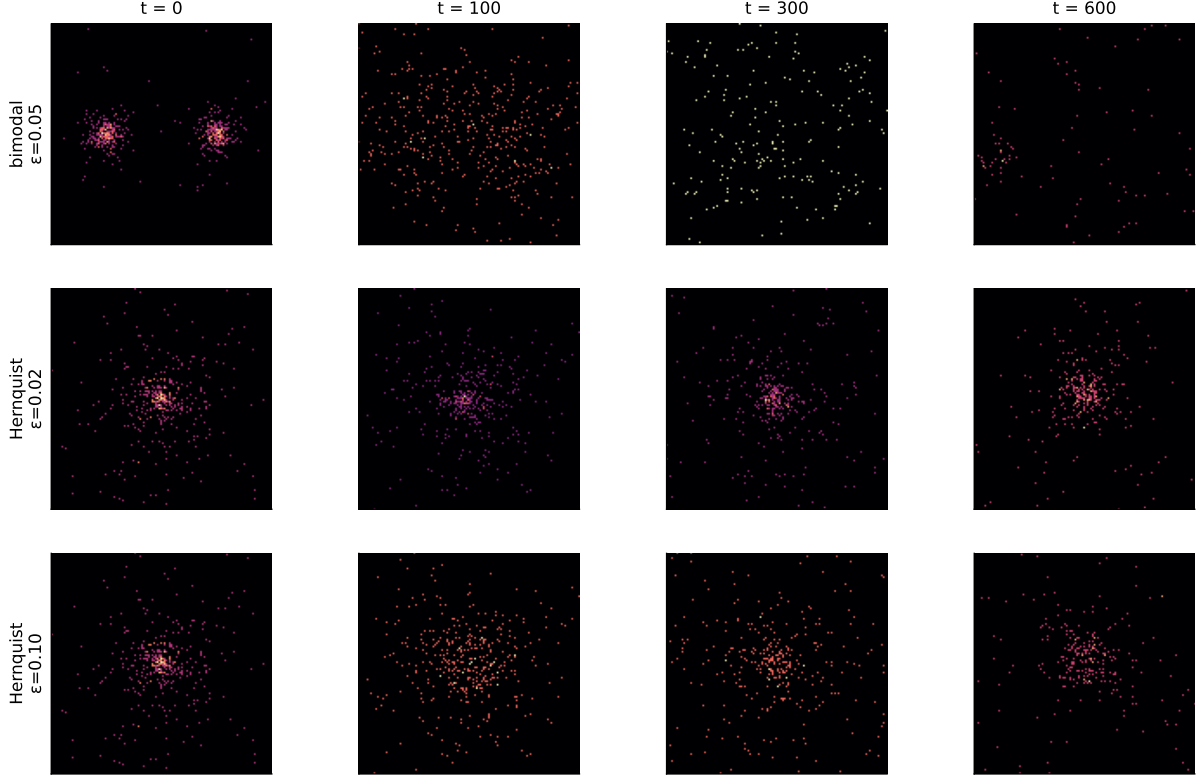


Figure 6: Time evolution: projected density (log scale, $N = 512$, direct-isolated). *Rows*: bimodal ($\varepsilon = 0.05$); Hernquist ($\varepsilon = 0.02$); Hernquist ($\varepsilon = 0.10$). *Columns*: $t = 0$, early, mid, final.

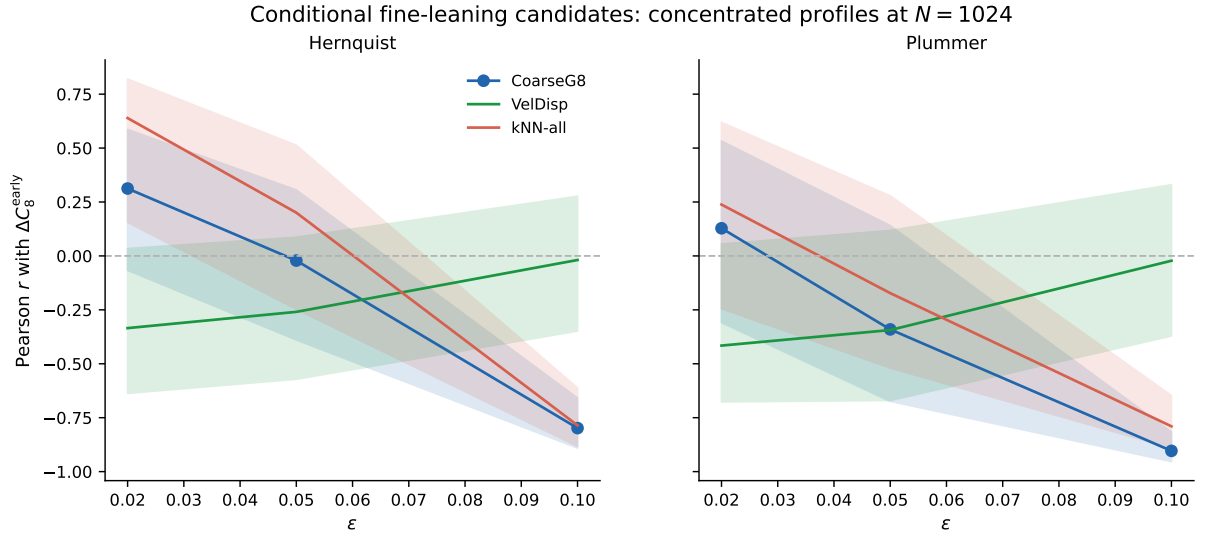


Figure 7: Observable class vs softening: $N = 1024$, direct-isolated. Solid: VelDisp (kinematic fine). Dashed: kNN-all (positional fine). Dotted: CoarseG8 (coarse). Shaded: 95% bootstrap CI. The VelDisp signal is present at $\varepsilon = 0.02$ for concentrated profiles and weakens with increasing ε . kNN-all does not outperform the coarse predictor in any cell tested.

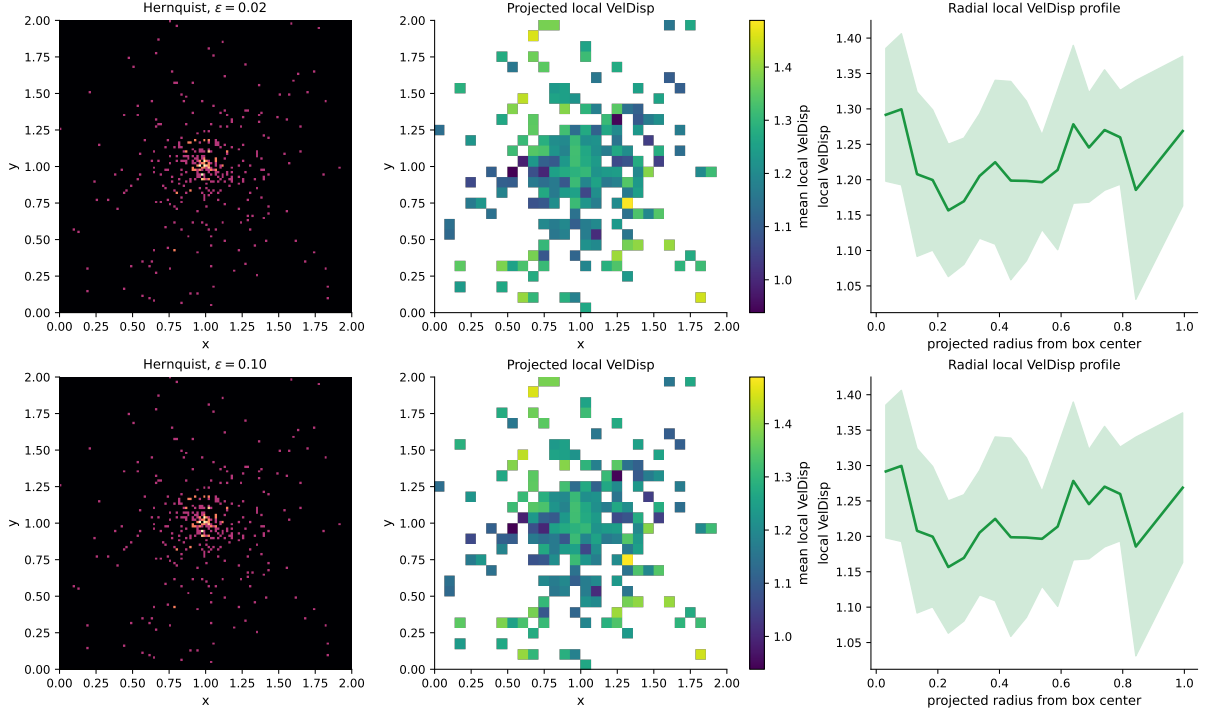


Figure 8: VelDisp mechanism: Hernquist profile, $N = 512$. Rows: $\varepsilon = 0.02$ (top); $\varepsilon = 0.10$ (bottom). Columns: projected density (log scale); local velocity-dispersion map; median $\sigma_v^{\text{loc}}(r)$ vs radius. At small softening the kinematic structure is concentrated toward the cusp; at large softening it is washed out.

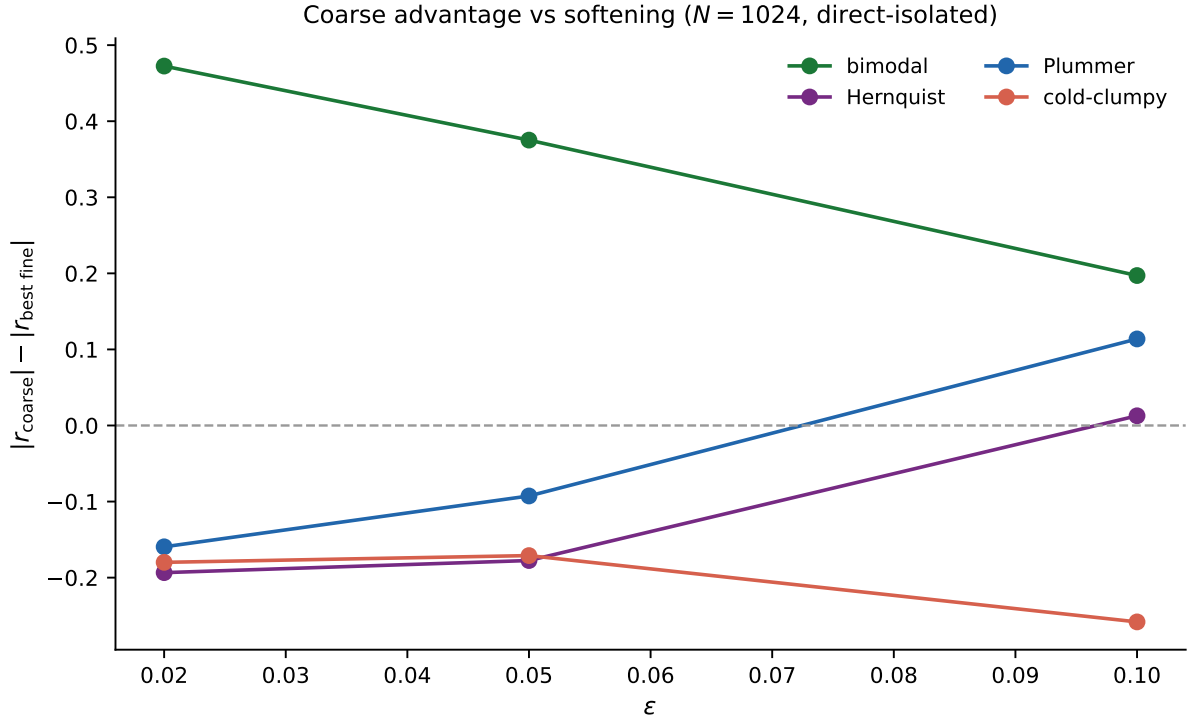


Figure 9: Coarse advantage vs softening: $N = 1024$, direct-isolated. Each curve shows $|r_{\text{coarse}}| - |r_{\text{best fine}}|$ as a function of ε for one IC family. Bimodal maintains a large positive advantage at all ε . Concentrated profiles show a suggestive fine-leaning signal at $\varepsilon = 0.02$ that weakens to tie or coarse-favoured by $\varepsilon = 0.10$.

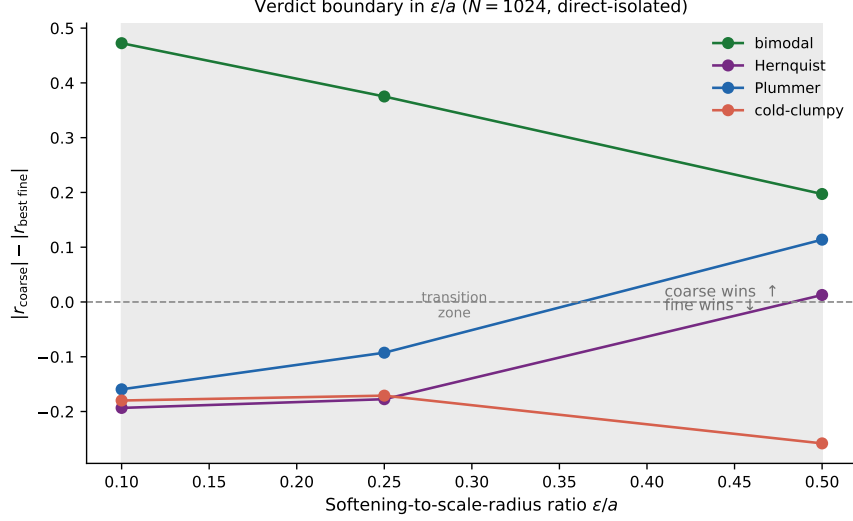


Figure 10: Verdict boundary in ε/a : coarse advantage $|r_{\text{coarse}}| - |r_{\text{best fine}}|$ vs the softening-to-scale-radius ratio, $N = 1024$, direct-isolated. Each point is annotated with the verdict (C = COARSE, F = FINE, T = tie). Bimodal stays above zero at all tested ε/a . Hernquist and Plummer show a suggestive fine-leaning signal at $\varepsilon/a \approx 0.10$ that shifts toward zero or positive between $\varepsilon/a \approx 0.10$ and 0.50 (shaded zone).

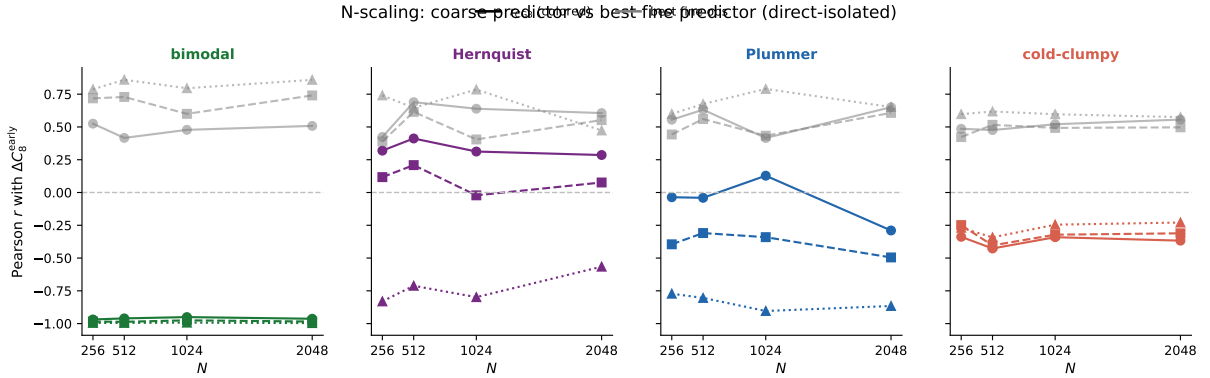


Figure 11: N-scaling at three softening values, direct-isolated. Each panel: one IC family. Coloured lines: r_{CG8} (solid/dashed/dotted for $\varepsilon = 0.02/0.05/0.10$). Grey: best fine predictor. Bimodal is stable near -0.98 at all N and ε .

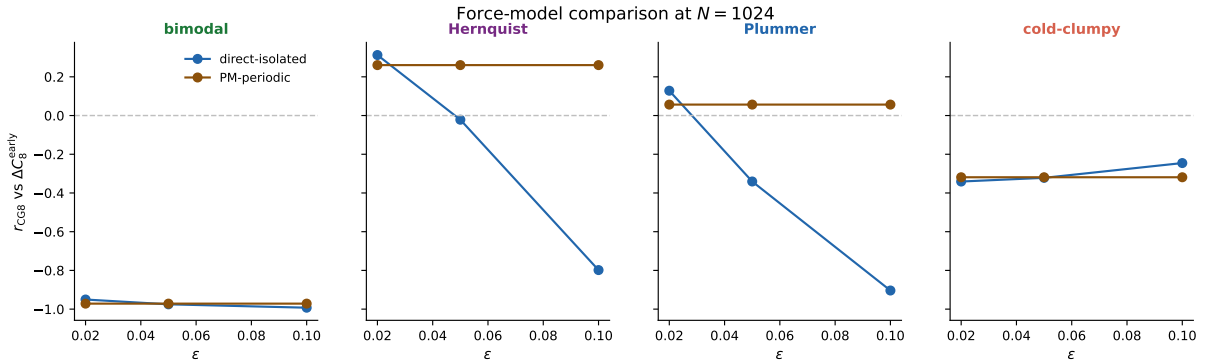


Figure 12: Force-model comparison: r_{CG8} vs ε , $N = 1024$. Blue: direct-isolated. Brown: PM-periodic. Shaded: 95% CI. Both models agree on the bimodal verdict. Concentrated profiles show quantitative differences but the same qualitative pattern.

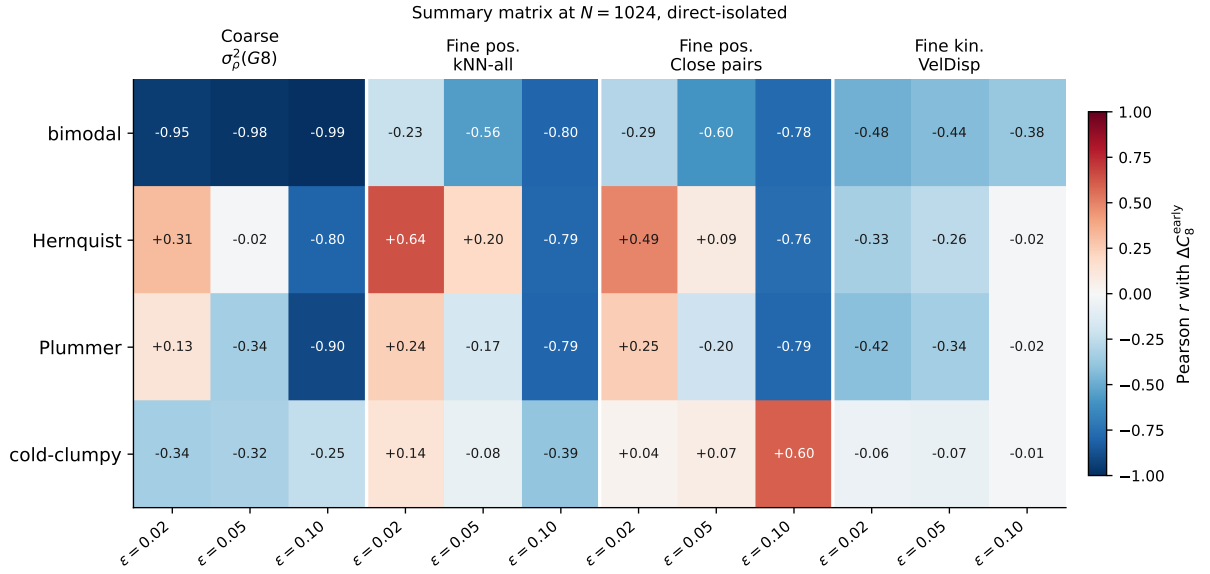


Figure 13: Summary matrix: Pearson r with $\Delta C_8^{\text{early}}$ by IC family (rows), observable class (groups), and ϵ (within-group columns), at $N = 1024$, direct-isolated. The coarse group (leftmost) shows strong negative r for bimodal at all ϵ . VelDisp (fine kinematic) shows the strongest fine signal for Hernquist and Plummer at $\epsilon = 0.02$. No tested positional fine observable outperforms the coarse predictor in any cell.

Data and Code Availability

All code available at https://github.com/kunalb541/Papers_Git

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